D-MOSFET

DC BIASING

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GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

50%

EXAM

50%

LABORATORY

PENALTY

0

ABSENT

3

45-min LATE

5 max

15-min LATE

TOPIC OUTLINE

D-MOSFET Construction

Regions of Operation

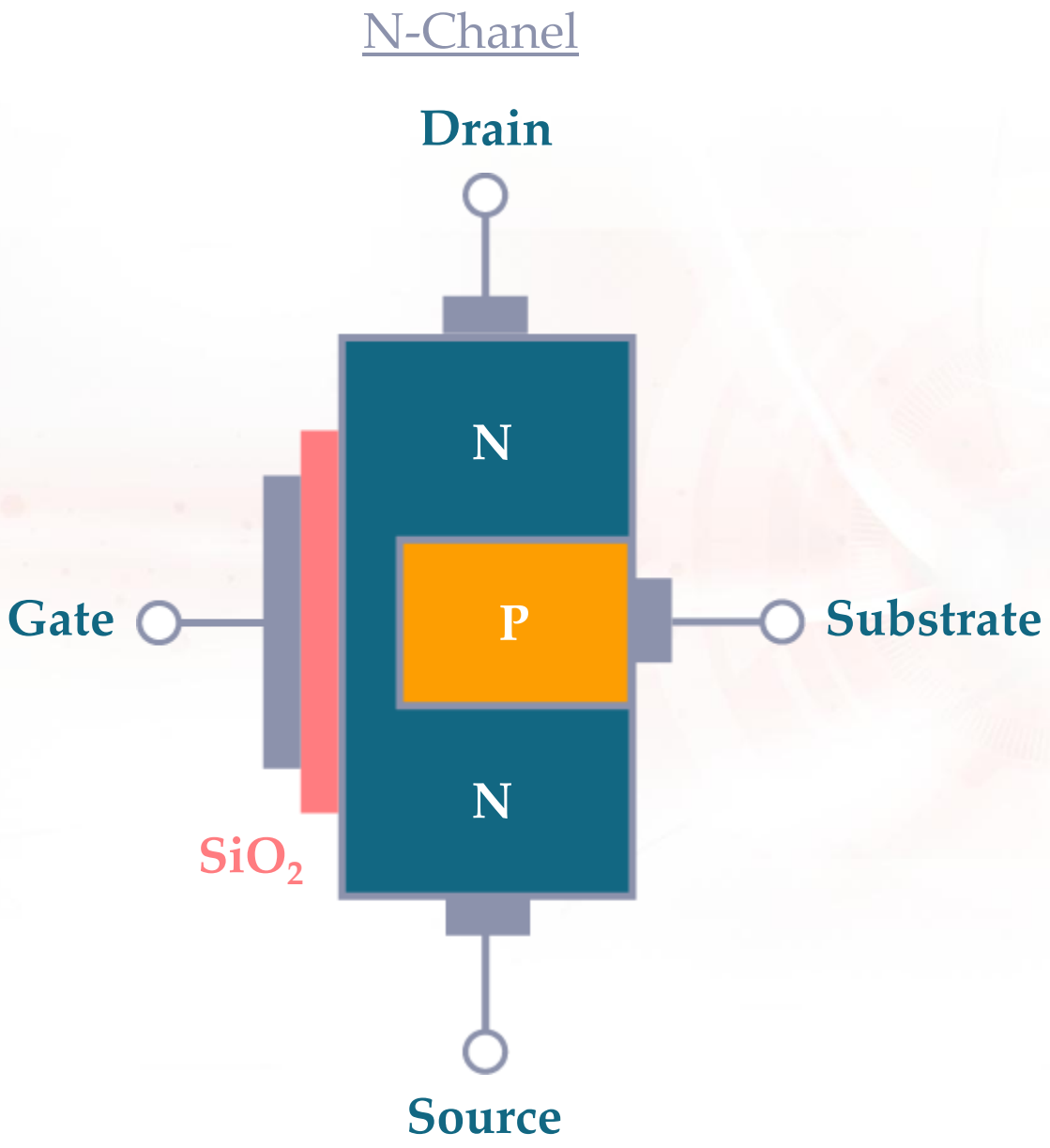
Transconductance Curve

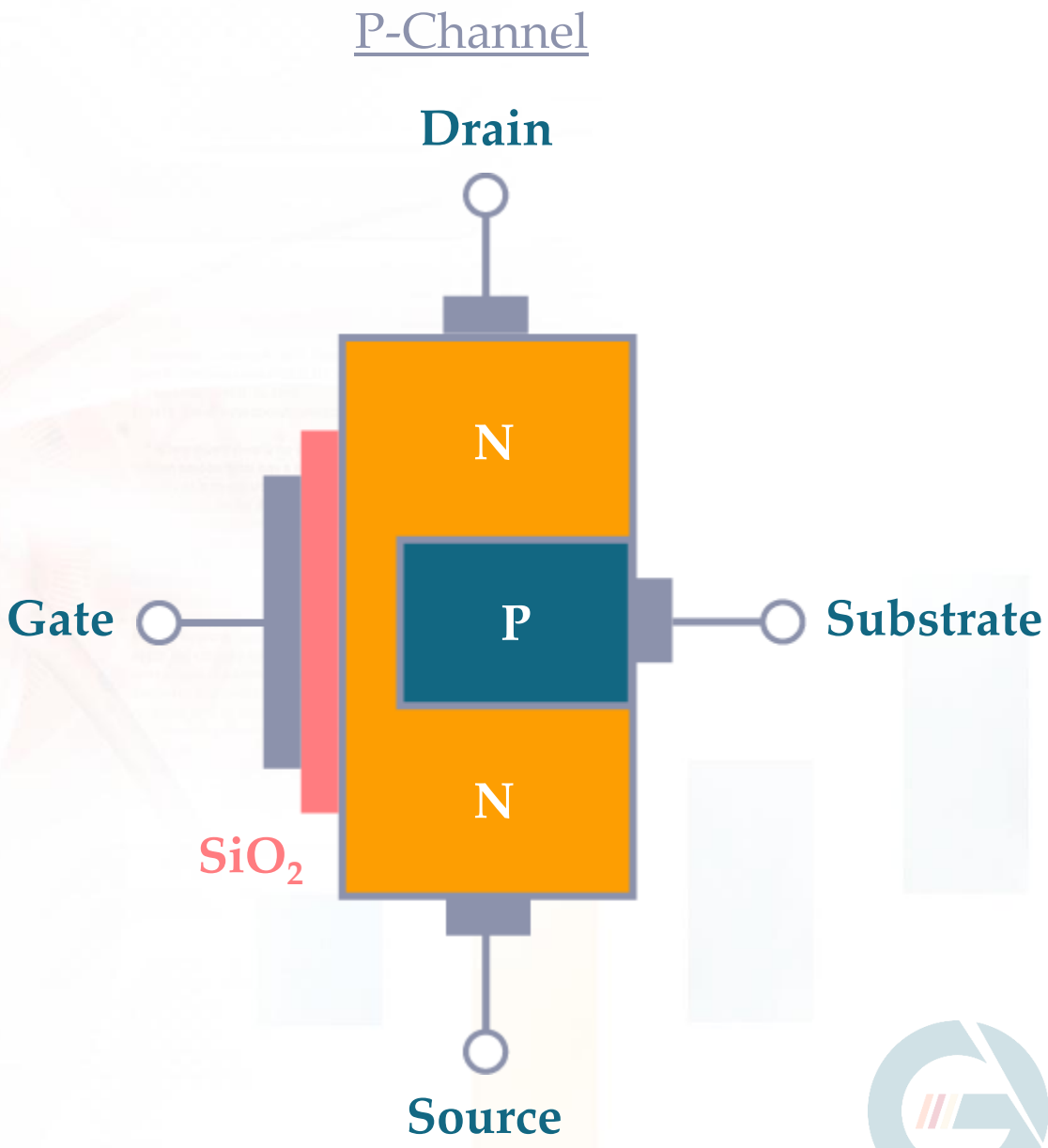
Self-Bias Configuration

Voltage-Divider Configuration

D-MOSFET CONSTRUCTION

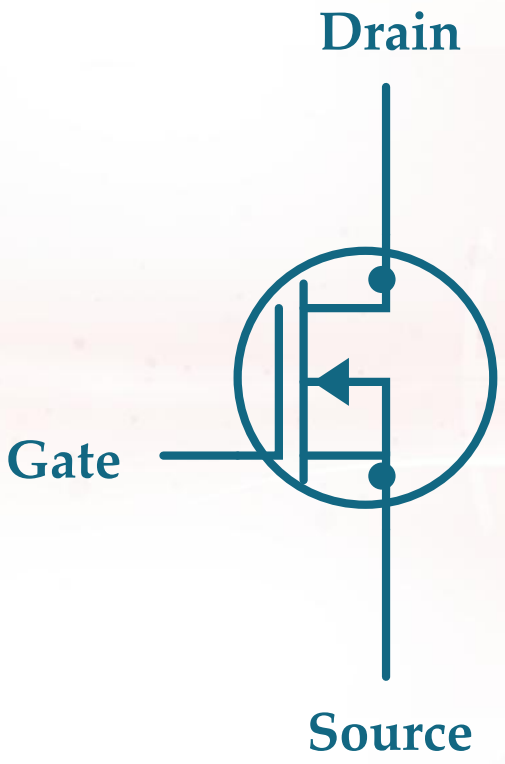
CONSTRUCTION



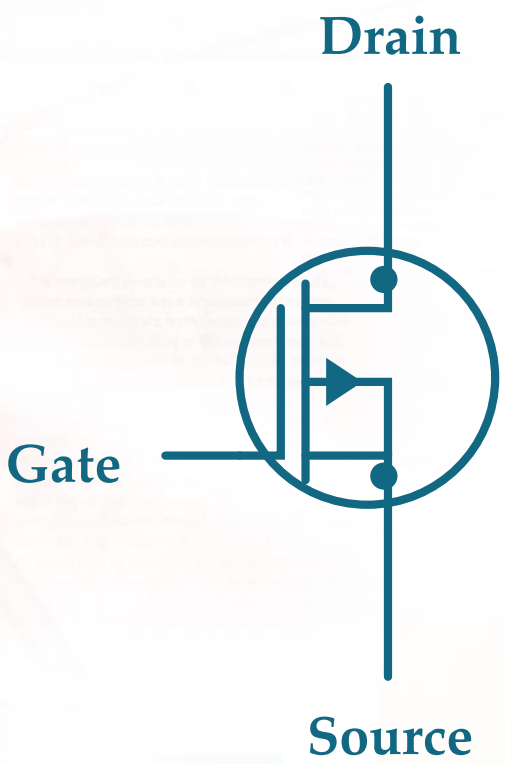


SCHEMATIC SYMBOL

N-Chanel

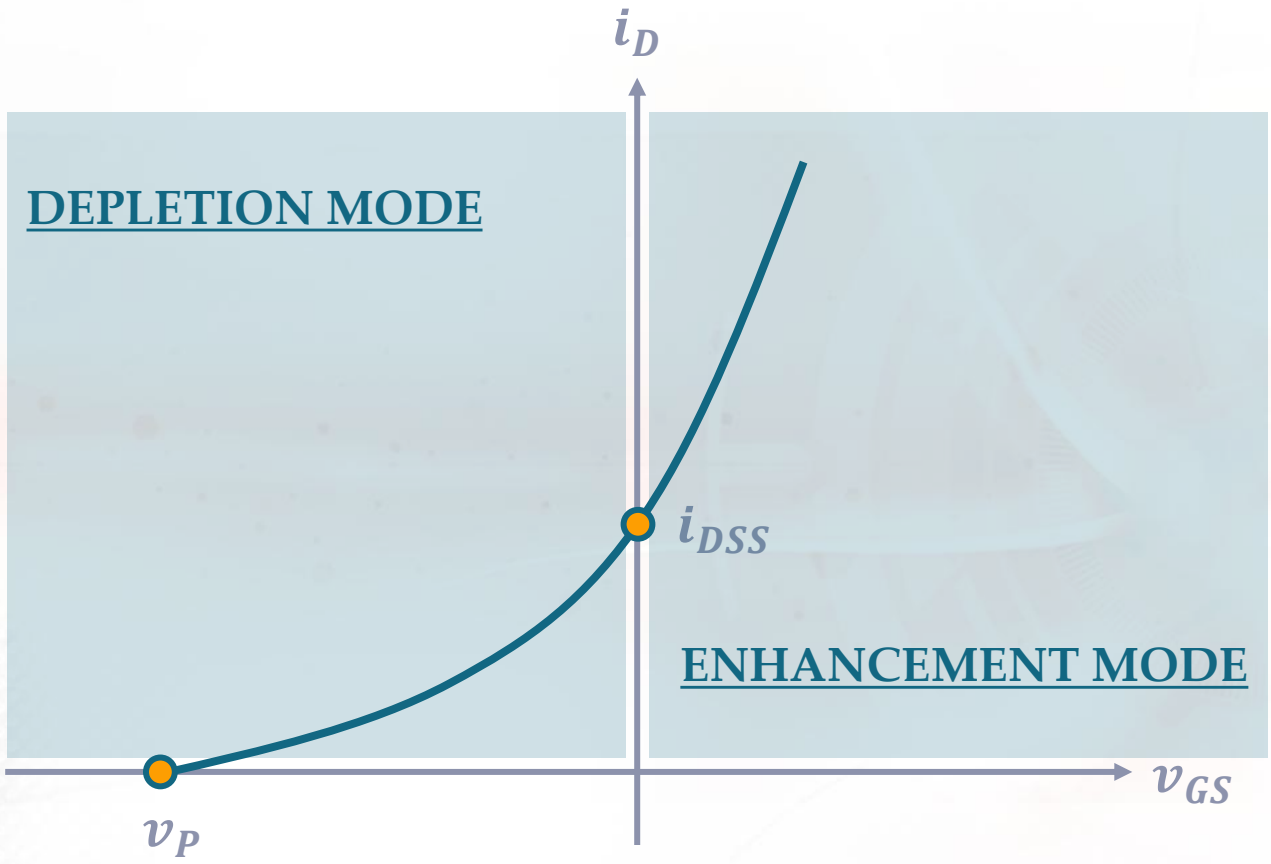


P-Channel



REGIONS OF OPERATION

DEPLETION-MODE MOSFET



MOSFET

Metal-oxide semiconductor FET

The primary difference between JFET and D-MOSFET is that D-MOSFETs permit operating points with **positive values** of v_{GS} and levels of i_D that **exceeds** i_{DSS}

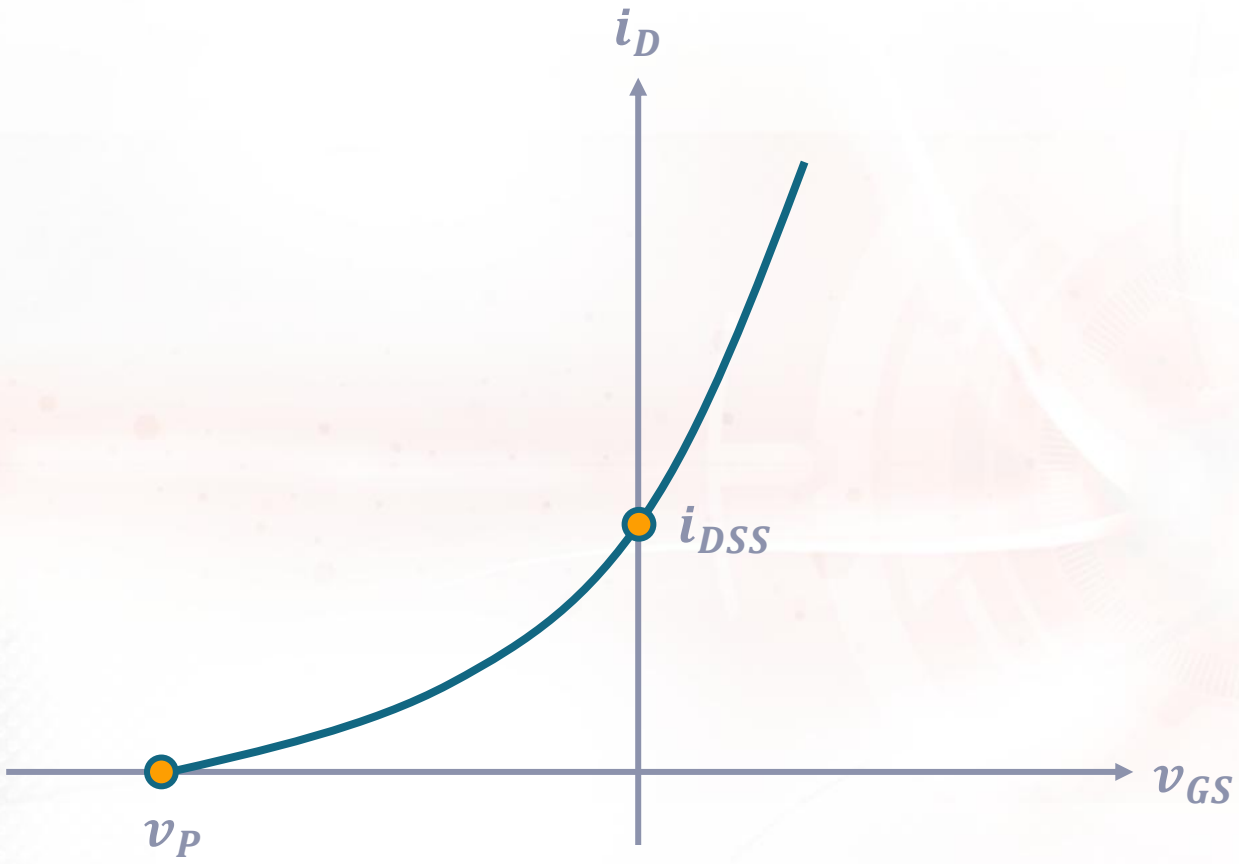
DRAIN CURVE

As with a JFET, when $v_{GS} = 0\ V$, the drain current will equal i_{DSS}

This demonstrates that the depletion-mode MOSFET is a **normally on** device

TRANSCONDUCTANCE CURVE

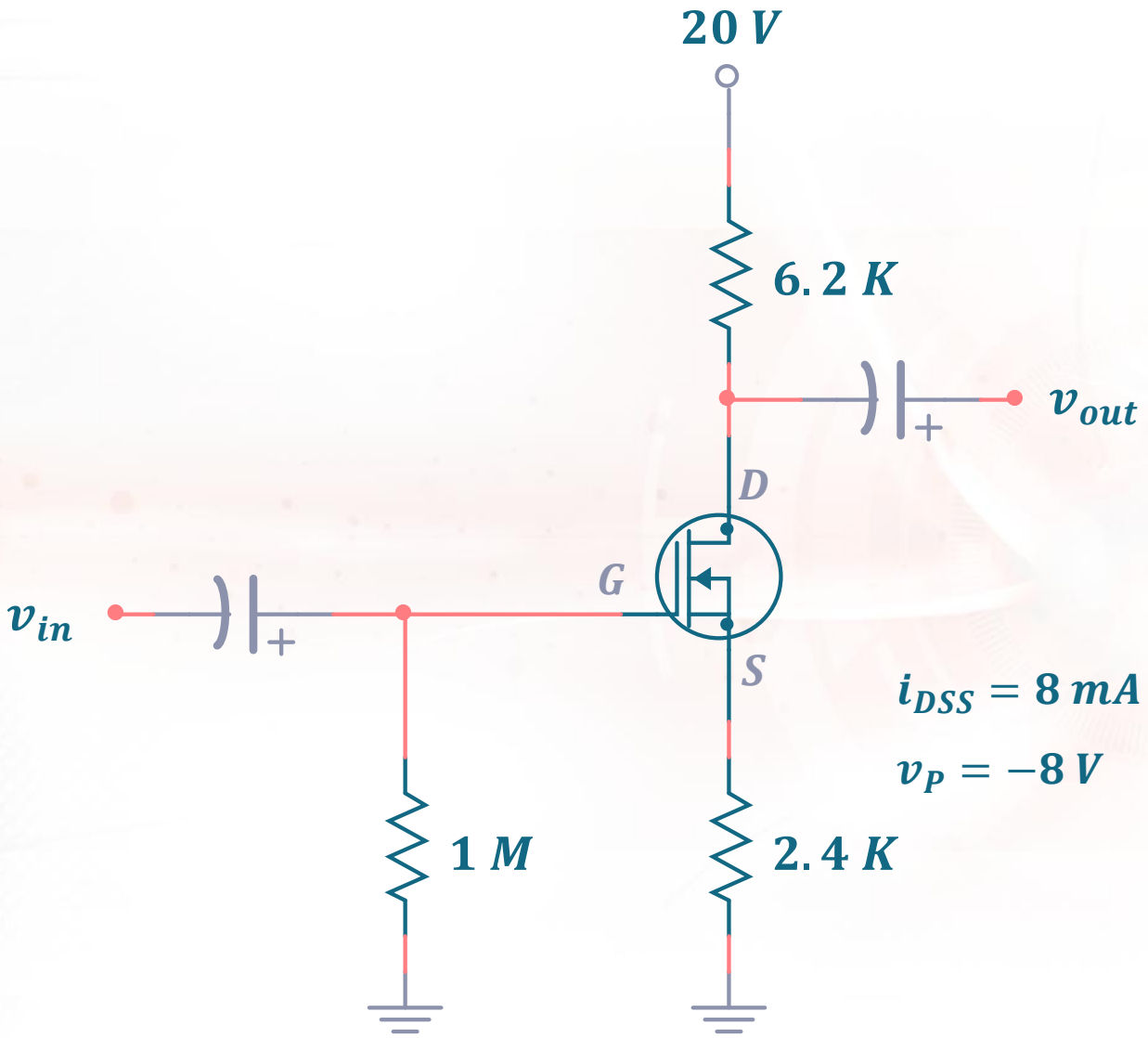
SHOCKLEY'S EQUATION



$$i_D = i_{DSS}\left(1 - \frac{v_{GS}}{v_P}\right)^2$$

Transconductance Curve

EXERCISE



$i_{DSS} = 8\ mA$

$v_P = -8\ V$

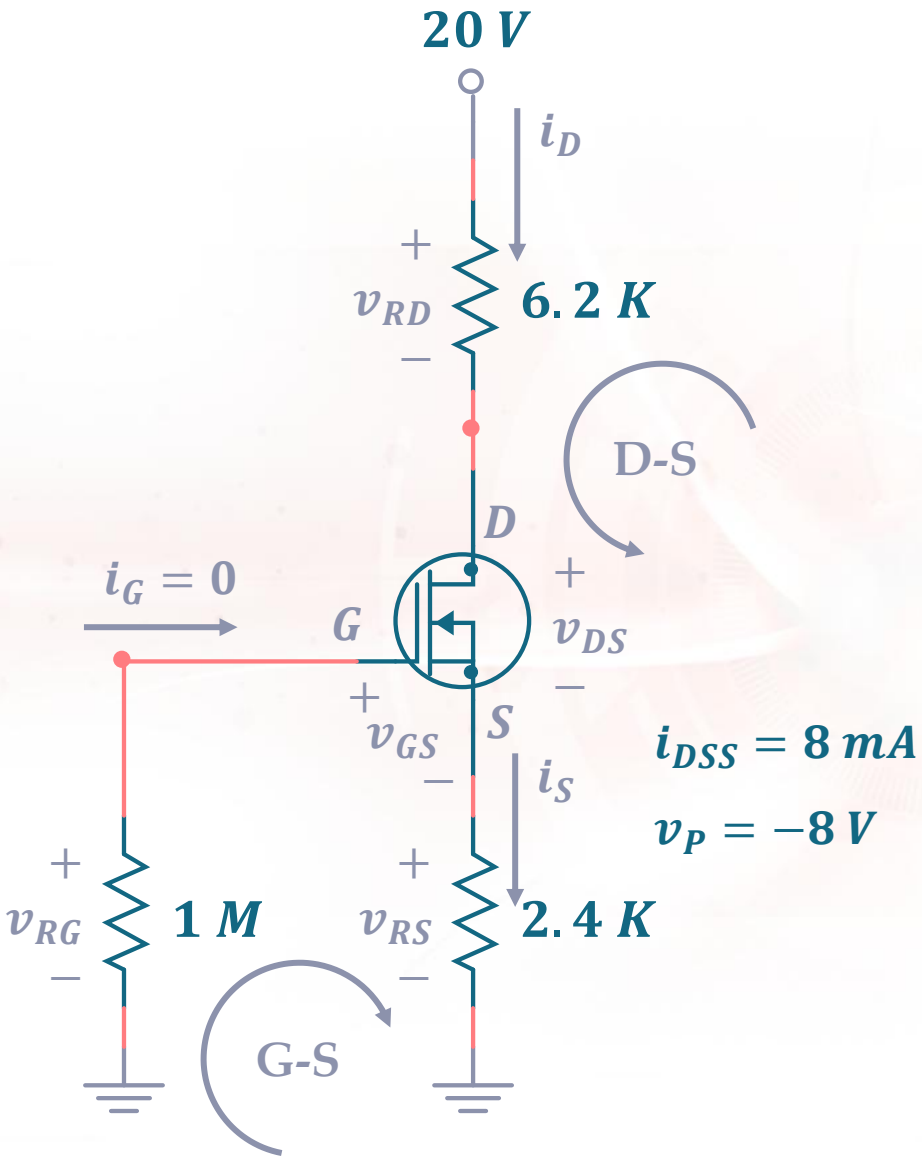
For the given network, determine the following:

v_{GSQ}

i_{DQ}

v_D

and **sketch** the transconductance curve.



KVL @ G-S

$$-v_{RG} + v_{GS} + v_{RS} = 0$$

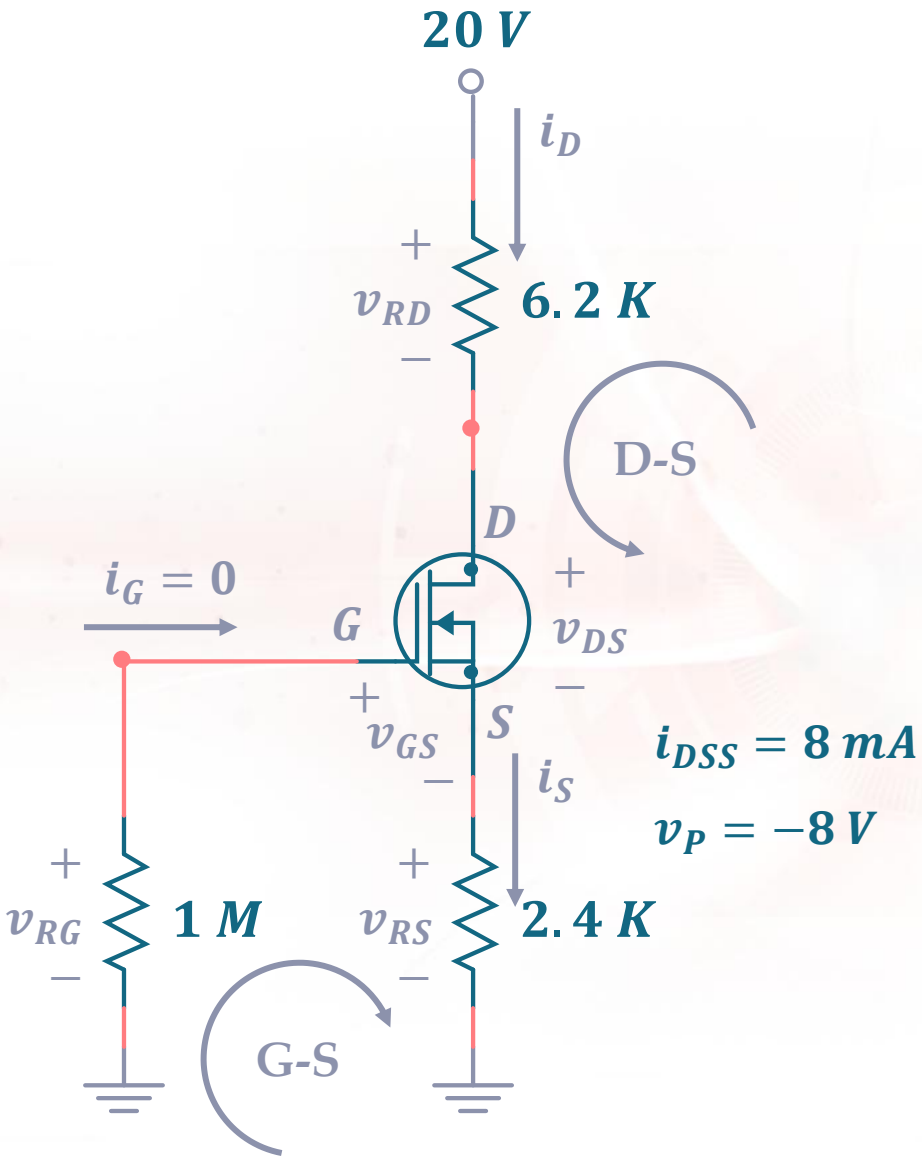
$$v_{GS} = -v_{RS} + v_{RG}$$

$$v_{GS} = -i_D R_S$$

$$v_{GS} = -i_{DSS}\left(1 - \frac{v_{GS}}{v_P}\right)^2 R_S$$

$$v_{GS} = -8\ mA\left(1 - \frac{v_{GS}}{-8}\right)^2 (2.4\ K)$$

$$v_{GS} = -19.2\left(1 + \frac{1}{4}v_{GS} + \frac{v_{GS}^2}{64}\right)$$



$i_{DSS} = 8\ mA$

$v_P = -8\ V$

KVL @ G-S

$$v_{GS} = -19.2\left(1 + \frac{1}{4}v_{GS} + \frac{v_{GS}^2}{64}\right)$$

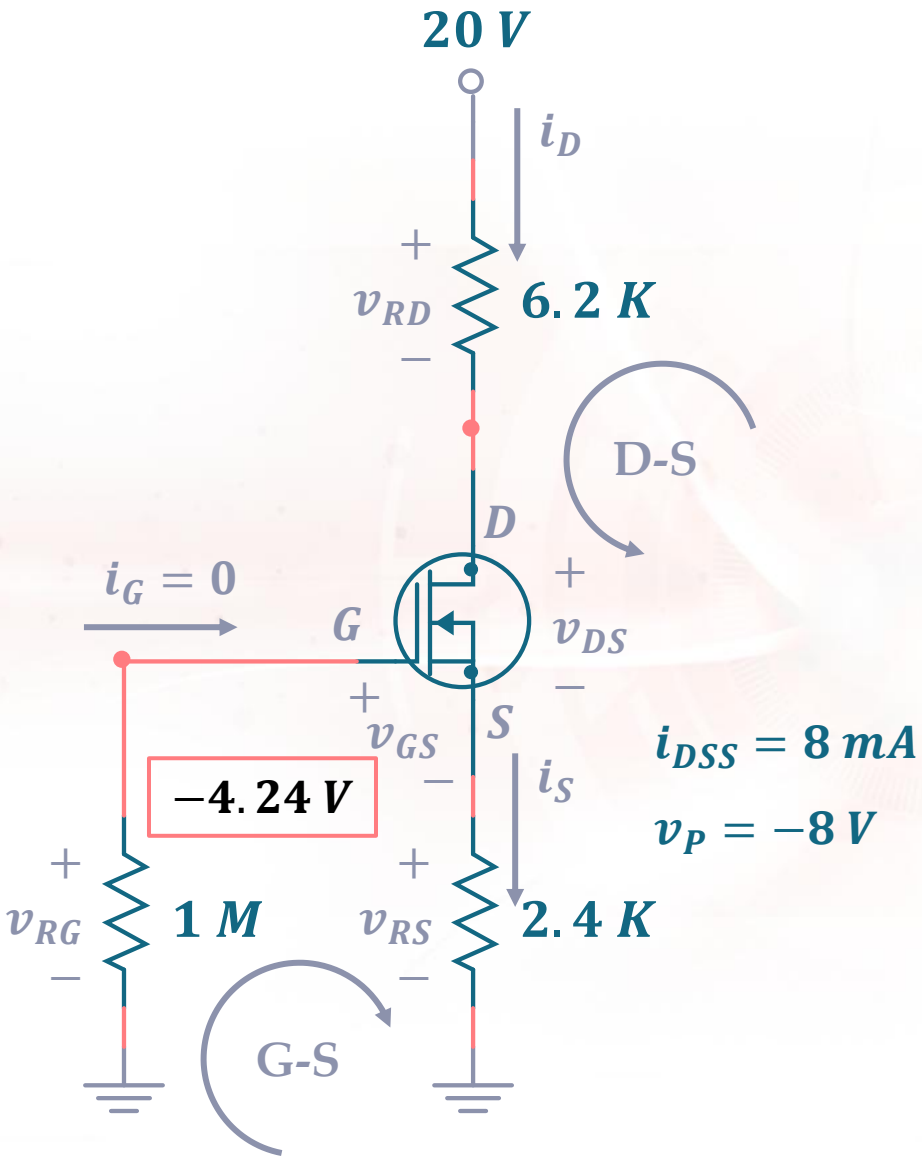
$$v_{GS} = -19.2 - 4.8v_{GS} - 0.3v_{GS}^2$$

$$0 = 19.2 + 5.8v_{GS} + 0.3v_{GS}^2$$

$$\boxed{v_{GSQ} = -4.24\ V}$$

ans

$$v_{GS} = -15.09\ V$$



$i_{DSS} = 8\ mA$

$v_P = -8\ V$

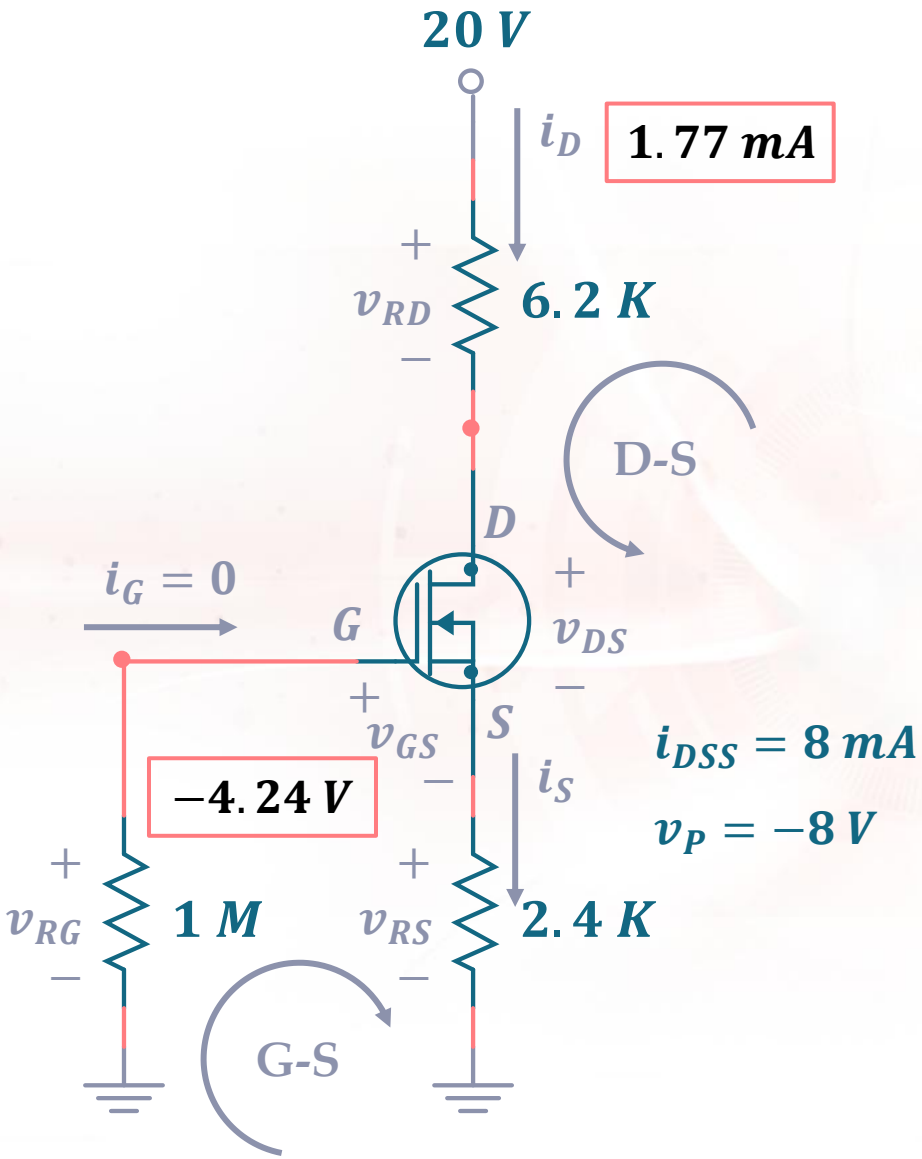
Shockley's Equation

$$i_D = i_{DSS}\left(1 - \frac{v_{GS}}{v_P}\right)^2$$

$$i_D = 8\ mA\left(1 - \frac{-4.24\ V}{-8\ V}\right)^2$$

$$\boxed{i_D = 1.77\ mA}$$

ans



$i_{DSS} = 8\ mA$

$v_P = -8\ V$

KVL @ D-S

$$-v_{DD} + v_{RD} + v_D = 0$$

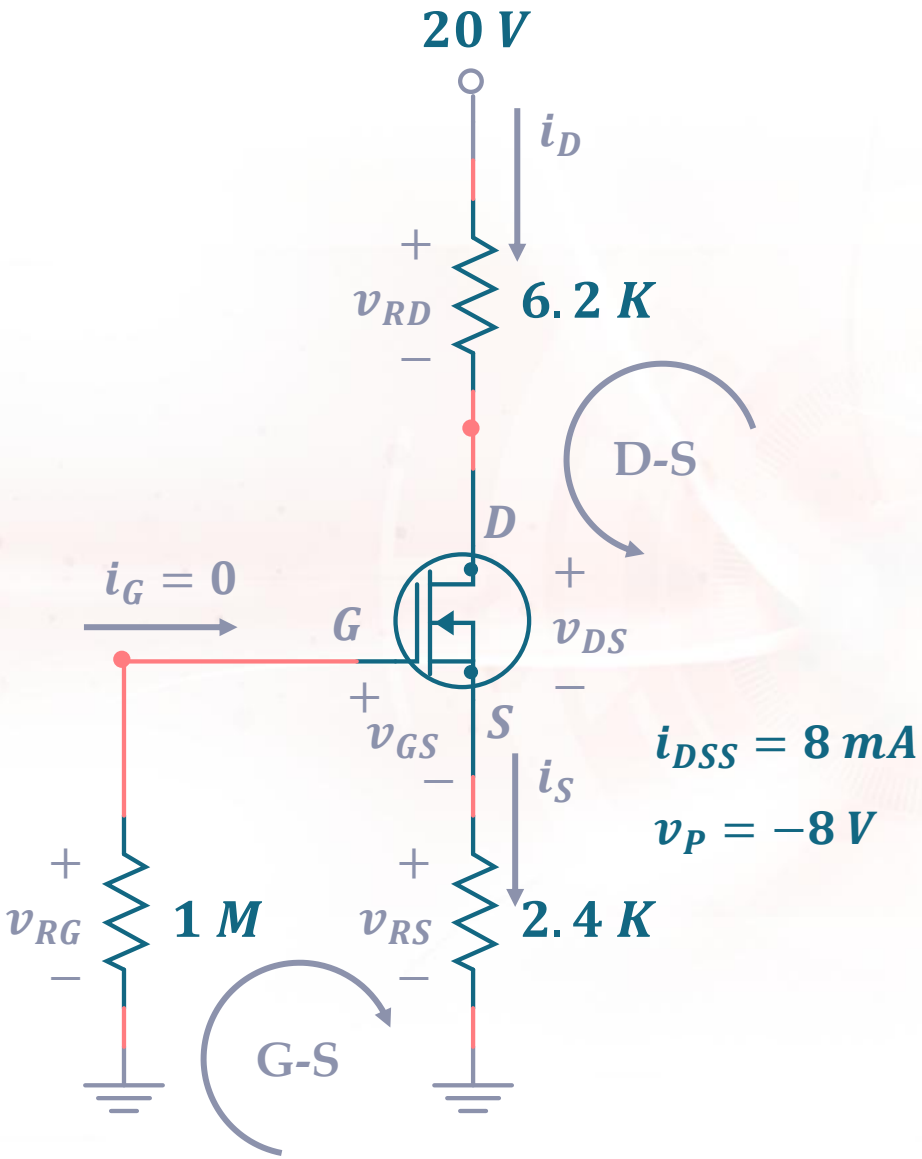
$$v_D = v_{DD} - v_{RD}$$

$$v_D = v_{DD} - i_D R_D$$

$$v_D = 20\ V - 1.77\ mA(6.2\ K)$$

$$\boxed{v_D = 9.03\ V}$$

ans



$i_{DSS} = 8\ mA$

$v_P = -8\ V$

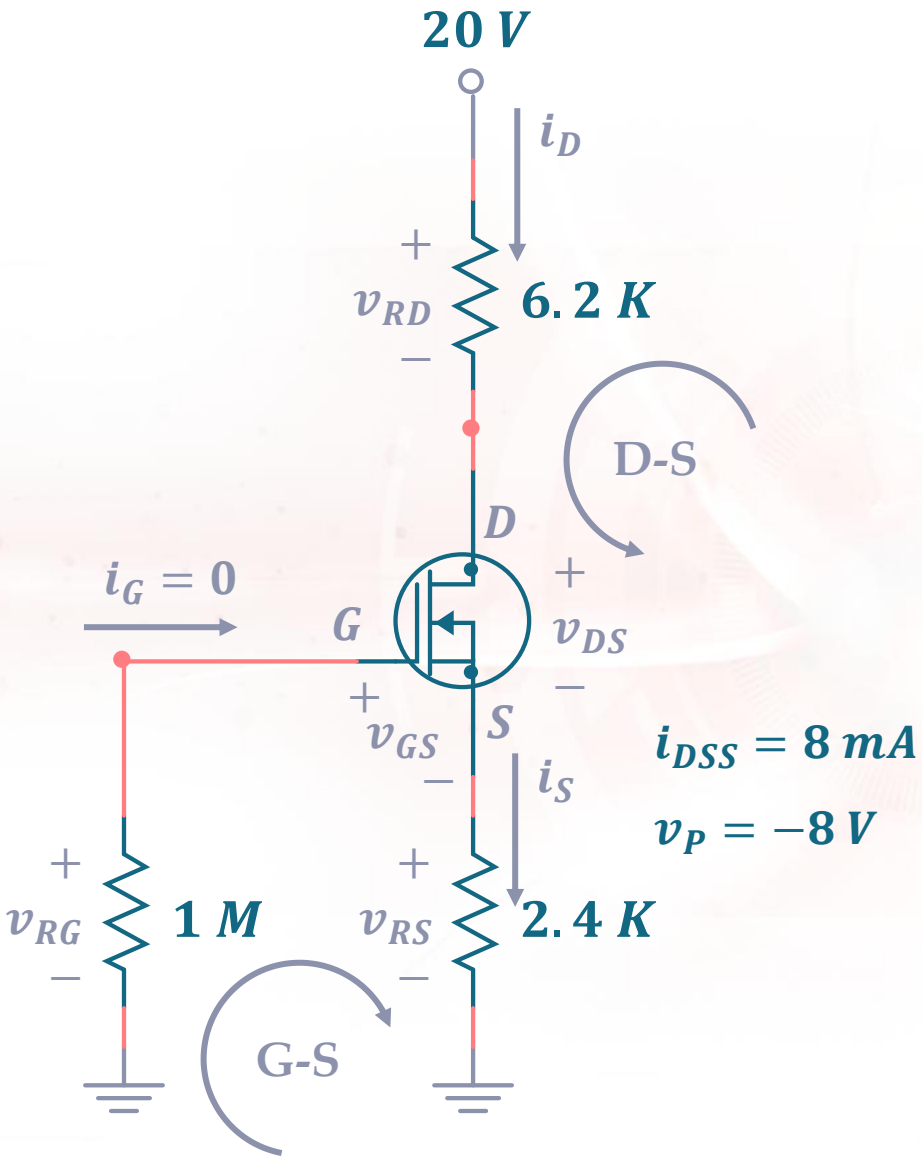
Transconductance Curve

$$i_D = \frac{1}{4} i_{DSS}\Big|_{v_{GS}=0.5v_P}$$

$$i_D = \frac{1}{4}(8\ mA)\Big|_{v_{GS}=-4\ V}$$

$$\boxed{i_D = 2\ mA\Big|_{v_{GS}=-4\ V}}$$

ans



$i_{DSS} = 8\ mA$

$v_P = -8\ V$

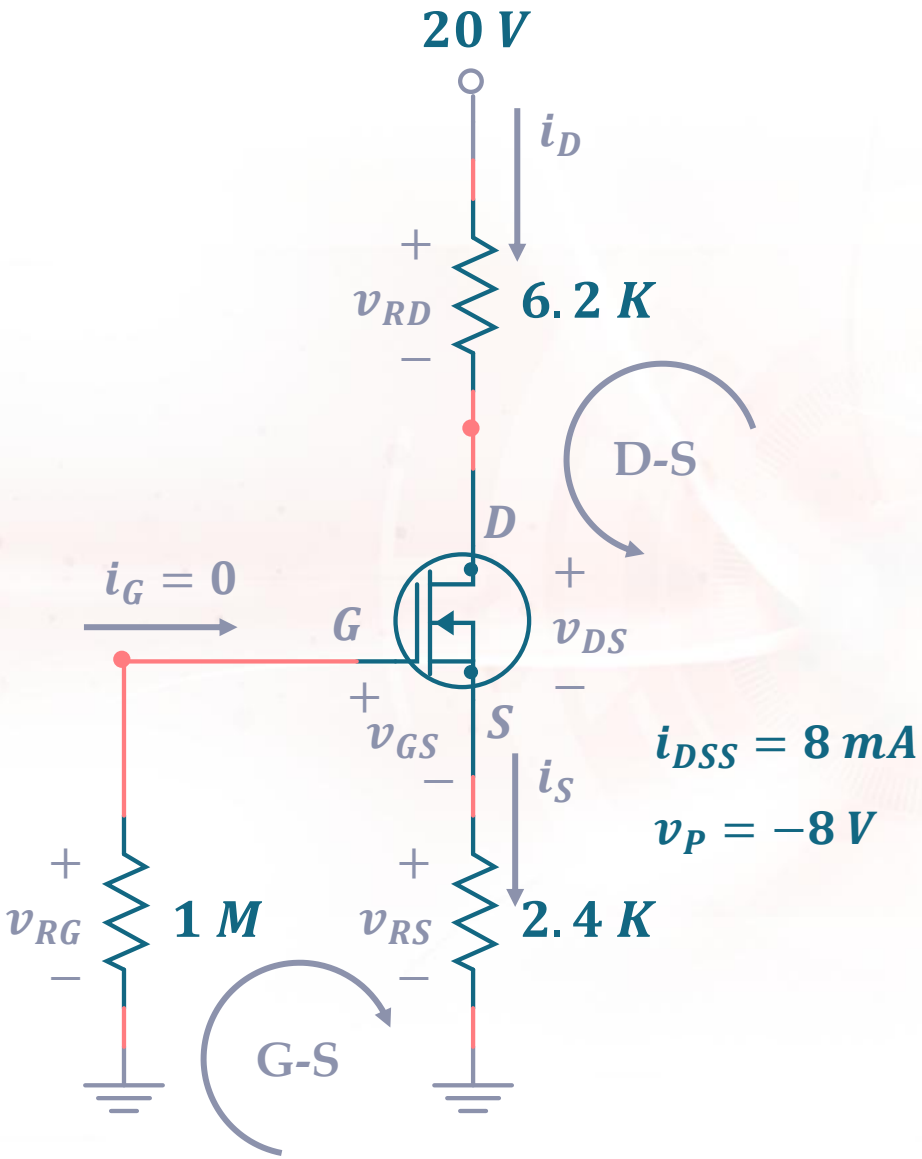
Transconductance Curve

$$v_{GS} = 0.3v_P\Big|_{i_d=0.5i_{DSS}}$$

$$v_{GS} = 0.3(-8)\Big|_{i_D=4\ mA}$$

$$\boxed{v_{GS} = -2.4\ V\Big|_{i_D=4\ mA}}$$

ans



$i_{DSS} = 8\ mA$

$v_P = -8\ V$

Transconductance Curve

$$i_D = i_{DSS}\left(1 - \frac{v_{GS}}{v_P}\right)^2\Bigg|_{v_{GS}=1\ V}$$

$$i_D = 8\ mA\left(1 - \frac{1\ V}{-8\ V}\right)^2\Bigg|_{v_{GS}=1\ V}$$

$$\boxed{i_D = 10.13\ mA\Big|_{v_{GS}=1\ V}}$$

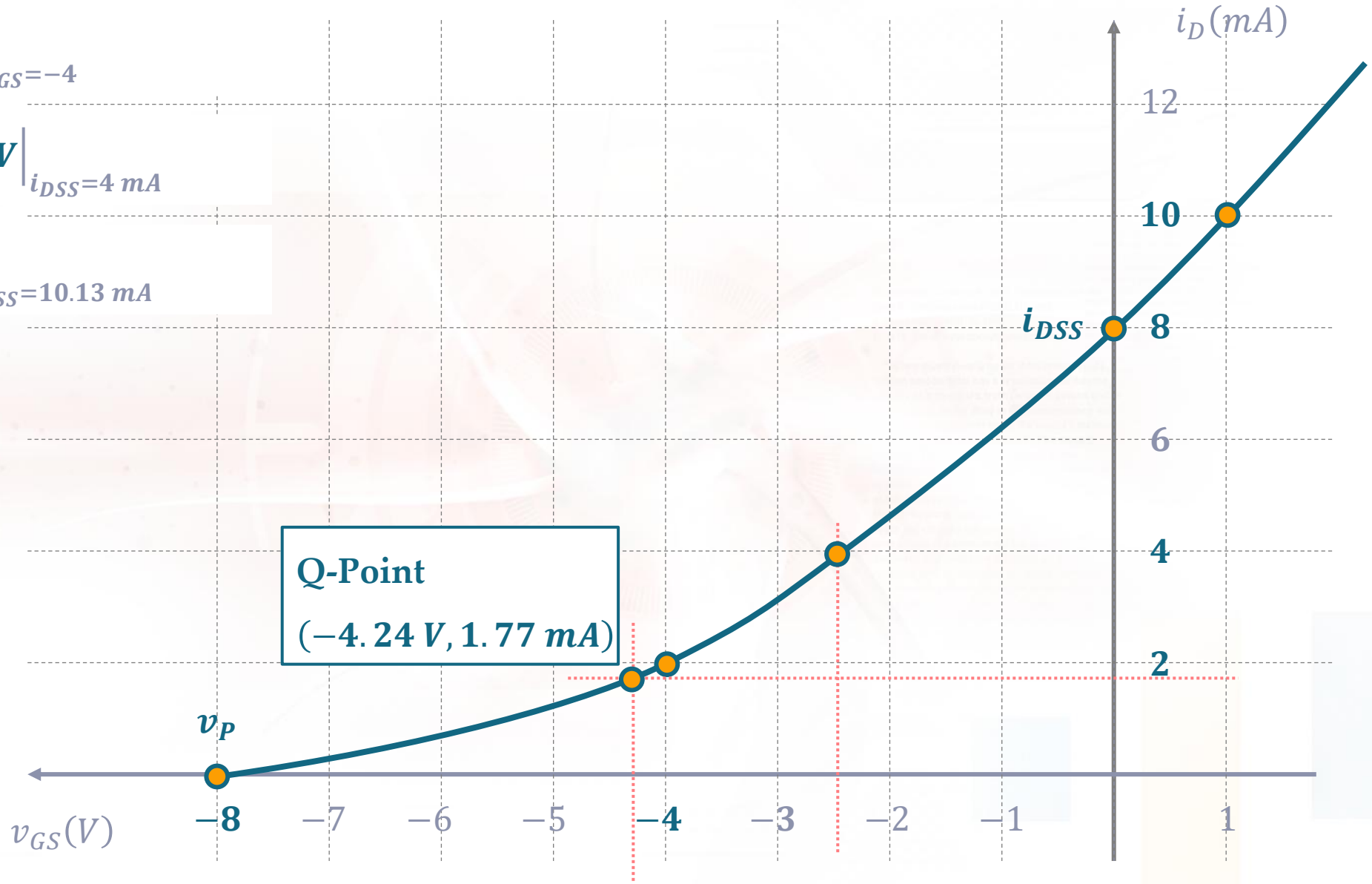
ans

TRANSCONDUCTANCE CURVE

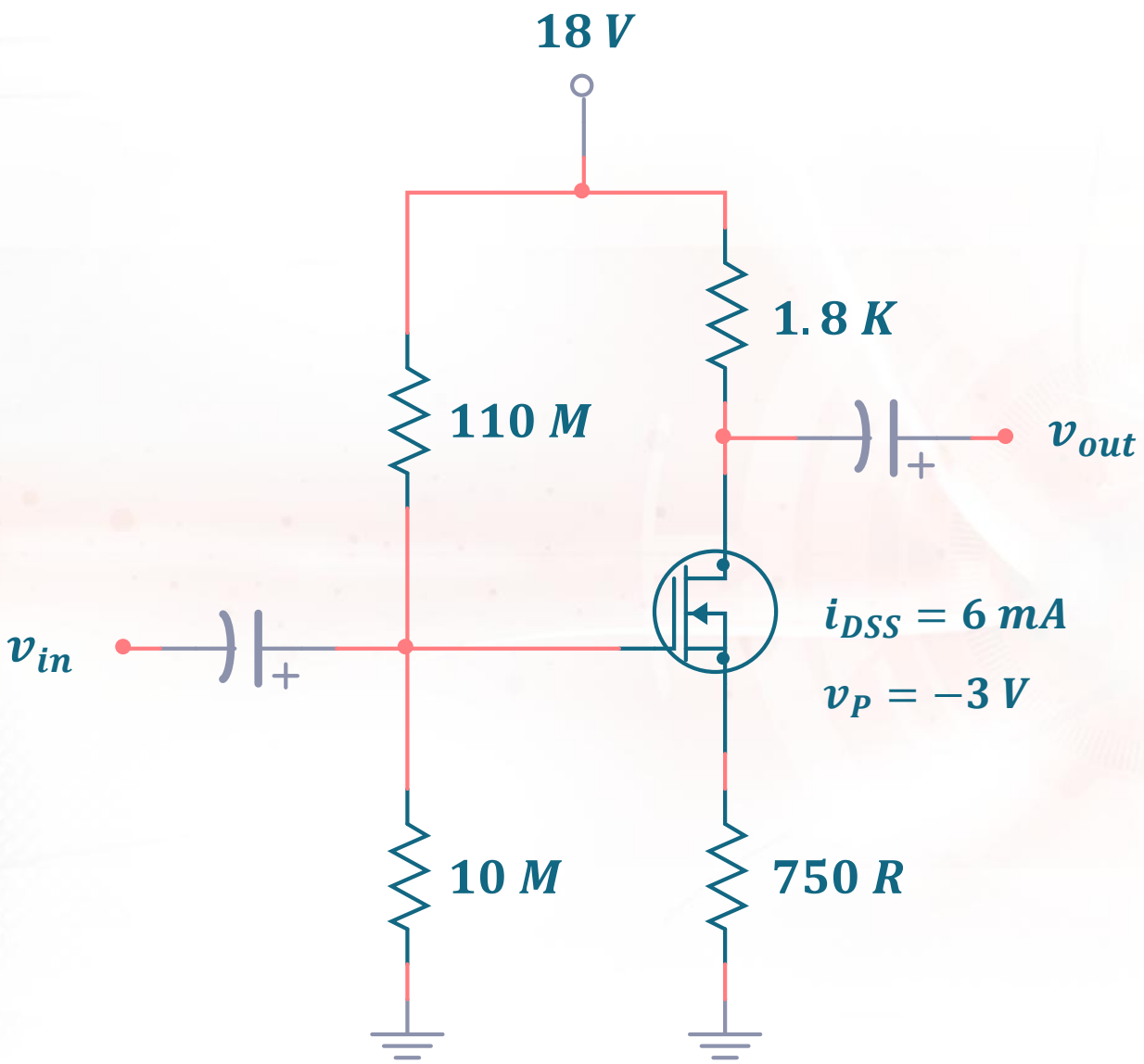
$$i_D = 2\ mA\Big|_{v_{GS}=-4}$$

$$v_{GS} = -2.4\ V\Big|_{i_{DSS}=4\ mA}$$

$$v_{GS} = 1\ V\Big|_{i_{DSS}=10.13\ mA}$$



EXERCISE



For the given network, determine the following:

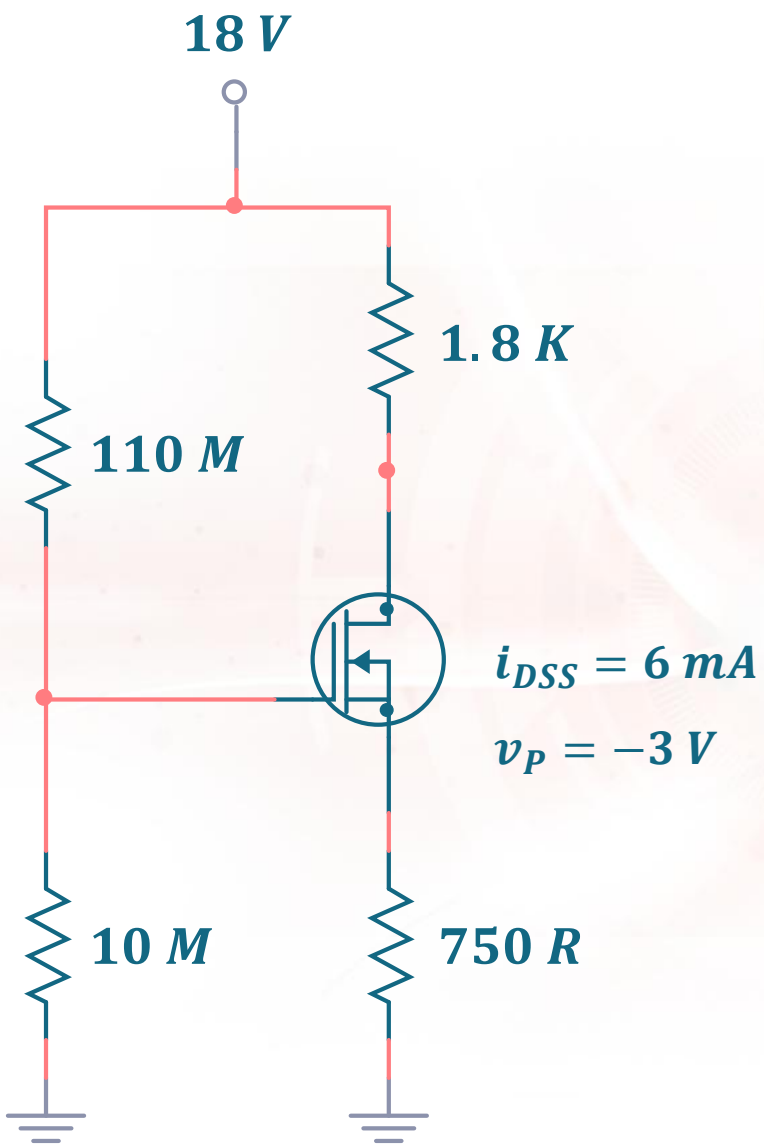
i_{DQ}

v_{GSQ}

v_{DS}

and **sketch** the transconductance curve.

EXERCISE



$i_{DSS} = 6\ mA$

$v_P = -3\ V$

LABORATORY